Other Gen Models, Deployment & Agents

Category : IMAGE GENERATION

Points : 100

Q1: What does the acronym GAN stand for?

A: Generative Adversarial Network

Category : IMAGE GENERATION

Points : 200

Q2: In a GAN, this paired network serves as an 'adaptive loss function' for the generator and is essentially thrown away after training.

A: The discriminator

Category : IMAGE GENERATION

Points : 300

Q3: To turn an unconditional GAN into one that can generate a specific '6' (not just any digit), you feed the generator a noise vector plus this extra input.

A: A class label (this yields a conditional GAN / cGAN)

Category : IMAGE GENERATION

Points : 400

Q4: This term in a VAE's loss penalizes the encoder when its learned 'clouds' drift from N(0,1), preventing them from collapsing to points or flying apart.

A: KL divergence (Kullback-Leibler divergence)

Category : **IMAGE GENERATION**

Points : **500**

Q5: At inference time, a Conditional VAE drops this component entirely - sampling z from N(0, I) and feeding it plus a text prompt to the decoder.

A: The encoder

Category : **POST-TRAINING**

Points : **100**

Q6: What does the acronym SFT stand for?

A: Supervised Fine-Tuning

Category : **POST-TRAINING**

Points : **200**

Q7: This parameter-efficient fine-tuning method freezes the base model and injects two small trainable matrices whose product approximates a full-size weight update.

A: LoRA (Low-Rank Adaptation)

Category : **POST-TRAINING**

Points : **300**

Q8: Unlike LoRA, this fine-tuning method prepends new trainable vectors to the Q, K, and V inputs inside the Transformer layers.

A: Prefix tuning

Category : **POST-TRAINING**

Points : **400**

Q9: In this post-training method, human annotators pick a preferred option between model outputs; with ~30-40k comparisons, the result is a model aligned to human taste.

A: RLHF (Reinforcement Learning from Human Feedback)

Category : POST-TRAINING

Points : 500

Q10: During SFT we average token-level cross-entropy over the label; the exponent of that average gives this metric - a common way to report language-model quality.

A: Perplexity

Category : AGENTIC AI

Points : 100

Q11: What does the acronym MCP stand for?

A: Model Context Protocol

Category : AGENTIC AI

Points : 200

Q12: Claude Code and Cursor are examples of this MCP component - sometimes also called a 'harness'.

A: MCP client

Category : AGENTIC AI

Points : 300

Q13: In MCP, this component is the counterpart to the client - it exposes tools and data (like a filesystem or an API) that the LLM can call.

A: MCP server

Category : AGENTIC AI

Points : 400

Q14: The core information-security risk with MCP agents: every tool result the agent reads - including sensitive files - ends up being sent here.

A: Sent to the LLM API (over the wire to the model provider)

Category : **AGENTIC AI**

Points : 500

Q15: Name any one of the four mitigation categories we discussed for securing agentic LLM pipelines.

A: Any of: contractual controls (enterprise API terms); self-hosted / VPC deployment; proxy or gateway with DLP / PII redaction; restricting tool access and permissions.

Category : **OTHER STUFF**

Points : 100

Q16: Name Google's text-to-image model that's named after a fruit.

A: Nano Banana ▯

Category : **OTHER STUFF**

Points : 200

Q17: What does the acronym RAG stand for?

A: Retrieval-Augmented Generation

Category : **OTHER STUFF**

Points : 300

Q18: This deployment trick rounds floating-point weights into integers - smaller memory footprint, faster inference, modest accuracy cost.

A: Quantization

Category : **OTHER STUFF**

Points : 400

Q19: In knowledge distillation, this smaller network is trained to mimic the outputs of a larger one - famously credited for DeepSeek's efficiency.

A: The student (model)

Category : OTHER STUFF

Points : 500

Q20: Our music Transformer's vocabulary is roughly this many tokens total - a tiny fraction of GPT's ~30K - covering Note On/Off, Time Shift, and Velocity events.

A: ~391 tokens (388 MIDI + 3 special)